Fresnel's biprism

1 Aim

- To setup the interference fringes of the double slit using a single source and Fresnels biprism.
- To determine the wavelength of sodium light using the Fresnels biprism.

2 Introduction

When two coherent beams of light of equal amplitude meet, an interference pattern consisting of a series of bright and dark fringes results (known as constructive and destructive interference, respectively).

Fresnels biprism is an example of an interferometric experiment depending on the division of wavefront to achieve interference of light from two coherent sources.

The biprism consists of a glass prism with two of its faces making an angle of nearly 180^o with each other, and both equally inclined to the third face, making the other two angles each equal to around $30''$. The Fresnel biprism is placed with its refracting edge parallel to the slit; this edge divides the incident light into two parts. When light from the source falls on the lower portion of the biprism, it is bent upwards and appears to come from the first virtual source. When light from the source falls on the upper portion of the biprism, it is bent downwards and appears to come from the second virtual source. The two virtual sources act as two coherent sources. Interference fringes of equal width (alternate bright and dark fringes) can be seen in the field of view of the eyepiece. The fringe width β is given by

$$\beta = \frac{D}{d}\lambda,$$

where d is the distance between the two virtual sources and D is the distance between the slit and the eye-piece and λ is the wavelength of the source.

The distance d cannot be measured directly since the images of the two slits are virtual. It is determined by placing a converging lens between the biprism and the

Figure 1: Apparatus in the lab.

screen and forming real images of the virtual slits on the screen; d can then be found using the lens magnification formula. The experiment proceeds in two parts:

- The fringes are setup and the fringe width is measured.
- The distance between the virtual sources is measured.

The optical bench has a heavy metal base with a metal scale along which four adjustable mounts can be moved, to carry a slit, a biprism, a lens and a micrometer eyepiece. A tangent screw can rotate the metal jaws clamping the slit and the biprism so that the slit and the edge of the biprism can be made exactly parallel to each other

3 Apparatus in the lab and procedure

Optical bench with lens mounts, sodium vapor lamp, Fresnel biprism, convex lens, slit, eyepiece with micrometer.

1. Adjust the mounts of the optical bench and check that the centre of the slit, the centre of the biprism and the centre of the eyepiece are all at the same height. Move the biprism close to the slit and first adjust the height of the prism and then the eyepiece till they are at the same height as the slit.

2. Switch on the sodium lamp and wait a while for it to heat up. Adjust the slit width to be narrow.

3. When you look at the biprism you should see two virtual light sources. Move your eye from one side of the biprism to the other, across the bench. One of the images will appear to cross the edge of the biprism from one side to the other. If the refracting edge of the biprism is exactly parallel to the slit, the image as a whole will appear to cross the edge.

4. The central edge of the biprism must be parallel to the slit. On looking through the eyepiece a bright patch of light should appear in the field of view. If you cannot see this bright patch, try rotating the top of the upright mount with the biprism about its vertical axis and move the mount along the bench till you see the bright patch.

5. Look through the eyepiece to see the fringes. If the fringes cannot be seen, move the biprism along the optical bench till the fringes appear in the field of view of the eyepiece. Rotate the prism in its own plane by the tangent screw till the fringes are seen clearly.

6. Fix the vertical crosswire of the eyepiece on the central bright fringe and move the eyepiece away from the biprism and see if there is any lateral (sideways) displacement of the fringes. If there is a lateral shift, move the biprism along the optical bench till the shift stops.

7. Measurement of fringe width β:
   Find the least count of the eyepiece micrometer. Adjust the distance of the eyepiece along the optical bench till the fringes are clear and distinct enough to be counted. Set the vertical crosswire on the centre of one of the bright fringes and note the main scale and micrometer readings. Set the crosswire on the next consecutive 20 fringes and note down the readings and determine the mean fringe width (if you cannot clearly distinguish 20 fringes, then try for the maximum number that you can clearly make out). Without disturbing the position of the slit and the biprism, change the position of the eyepiece and take two more sets of readings to measure the fringe width.

8. Measure the distance D between the slit and the eyepiece from the attached scale on the optical bench.

9. Measurement of d:
   Place the convex lens between the prism and eyepiece, close to the eyepiece. Move the lens towards the biprism till you see two sharp images of the slit. Measure the distance d between the two images of the slit using the eyepiece micrometer. Fix the vertical crosswire on one image and note the micrometer reading and move

the screw till the crosswire falls on the second image. The difference between the two readings gives the distance d_1. Again move the lens toward the biprism till another set of two slit images are seen and measure the distance d_2 between the second set of images; $d = \sqrt{d_1 d_2}$. Take a set of readings (when lens is near the slit and when lens is near the eyepiece) and find the mean d.

10. Compute the wavelength of the sodium light and compare your result with the standard value.

3.1 Precautions

1. The Fresnel biprism should be clean and should have no finger prints.
2. The refracting edge of the biprism should be parallel to the slit.
3. Avoid backlash error and turn the micrometer screw in one direction only while taking readings.

3.2 Points to Ponder

1. How would the fringes change is the Fresnel biprism is made thinner (or thicker)?
2. Why is the light from the sodium lamp pink when it is switched on and turns yellow later?
3. What will be the result if the experiment is performed under water?